Data Communication & Computer Networks (CN)

Topic 4 of 7 : Networking Devices

53 Pending Questions — staged from repo exam PDFs & trusted sources — **NEED VERIFICATION** (answers + Hindi to be verified during topic build)

Reference Book: **Data Communications and Networking — Behrouz A. Forouzan**

Prepared for: **RPSC Programmer • ACP • RSMSSB IA • SA • DD • BCI • SCI • REET • RAS • DSSSB • KVS • NVS • HTET • SSC CGL/CHSL • RRB • IBPS • UPSC & all Rajasthan + Central Govt. Exams (Computer Awareness)**

Total Questions: **53** pending questions (staging)

Question Type: MCQ — 4 options (A, B, C, D) per question, in English + Hindi

Layout: Left = **question** | Right = **explanation** (green): ✔ **why the correct answer is right** + ★ **Nukta** (key exam point)

Language rule: Hindi is the second language; **technical terms stay in English** (topology, protocol, jitter…)

Verification: Every answer cross-verified **4×** (Forouzan + official answer keys of Subject Test / Unit Test Solutions). Wrong options verified internally.

Syllabus Coverage — sub-topic wise order

How to read: Left side: question + 4 options (try it yourself first). Right side (green): ✔ correct answer with why it is right, and ★ **Nukta** — the one-line exam tip to remember. Practice in 2 passes: first read question + options only, then check the green column.

System: generated by **tools/cn_build_topic.py** from **tools/data/topic4_data.py**. To add more questions later: drop PDFs into **Computer Network/Incoming/** and run **tools/cn_pipeline.py** — new questions auto-classify topic-wise. See **Computer Network/README.md**.

Pending Review — auto-ingested, needs verification

Auto-extracted & classified; answers to be verified before counting.

P1. Which device reduces network collision? collision

(A) Hub
(B) Switch
(C) Repeater
(D) Modem

[Source: BCI Subject Test-8.pdf | auto-classified: Topic 4 / Hub / Repeater / Modem]

⚠ auto-ingested — answer needs verification

P2. Which device creates a separate broadcast domain? broadcast domain **(A)** Switch **(B)** Bridge **(C)** Router **(D)** Repeater [Source: BCI Subject Test-8.pdf | auto-classified: Topic 4 / Router / Brouter]	⚠ auto-ingested — answer needs verification
P3. Which device stores MAC addresses in a CAM table? - MAC addresses CAM **(A)** Hub **(B)** Router **(C)** Switch **(D)** Modem [Source: BCI Subject Test-8.pdf | auto-classified: Topic 4 / Hub / Repeater / Modem]	⚠ auto-ingested — answer needs verification
P4. A switch receives a frame with a destination MAC address not present in its MAC table. The switch is in a network connected to a router that separates broadcast domains. What will the switch do? MAC address MAC broadcast domains **(A)** Send frame to router only **(B)** Flood in the same broadcast domain **(C)** Drop the frame **(D)** Broadcast in entire network [Source: BCI Subject Test-8.pdf | auto-classified: Topic 4 / Switch / NIC / Bridge]	⚠ auto-ingested — answer needs verification
P5. Router is connected between two switches. If a broadcast frame is sent in Switch-A, what will Switch-B receive? Switch A broadcast , Switch-B **(A)** Broadcast **(B)** Unicast **(C)** Nothing **(D)** Multicast [Source: BCI Subject Test-8.pdf | auto-classified: Topic 4 / Switch / NIC / Bridge]	⚠ auto-ingested — answer needs verification
P6. Which device will break collision domains but NOT broadcast domains? collision domain broadcast domain **(A)** Switch **(B)** Router **(C)** Hub **(D)** Gateway [Source: BCI Subject Test-8.pdf | auto-classified: Topic 4 / Hub / Repeater / Modem]	⚠ auto-ingested — answer needs verification

P7. A network uses VLANs and a router-on- a-stick configuration. Which device handles VLAN tagging and which device handles inter-VLAN routing? VLAN router-on-a-stick VLAN - inter VLAN routing -

(A) Switch – VLAN tags, Router – routing
(B) Router – VLAN tags, Switch – routing
(C) Both Switch and Router do VLAN tagging
(D) None do VLAN tagging

[Source: BCI Subject Test-8.pdf | auto-classified: Topic 4 / Router / Brouter]

⚠ auto-ingested — answer needs verification

P8. A packet travels from a wired PC → Switch → Access Point → Wireless laptop. Which devices operate at Layer-2 only? wired PC → Switch → Access Point → Wireless laptop Layer-2

(A) Switch only
(B) Access Point only
(C) Both Switch and Access Point
(D) None

[Source: BCI Subject Test-8.pdf | auto-classified: Topic 4 / Switch / NIC / Bridge]

⚠ auto-ingested — answer needs verification

P9. A switch has 4 VLANs configured. How many broadcast domains exist? 4 VLAN broadcast domain

(A) 1
(B) 2
(C) 4
(D) Depends on router

[Source: BCI Subject Test-8.pdf | auto-classified: Topic 4 / Router / Brouter]

⚠ auto-ingested — answer needs verification

P10. @A B A B Perfect Match B

(A) Packet Filtering 1.Transport Layer
(B) Stateful Inspection 2.Network Layer
(C) Circuit Level 3. Application Gateway Layer Security
(D) Stateful Inspection 4. Session Layer Firewall (a) a-1, b-2, c-3, d-4 (b) a-2, b-1, c-4, d-3 (c) a-4, b-3, c-2, d-1 (d) a-3, b-1, c-4, d-2

[Source: BCI Subject Test-9.pdf | auto-classified: Topic 4 / Gateway / Firewall]

⚠ auto-ingested — answer needs verification

P11. Which type of firewall inspects packets at the network layer and decides based on IP addresses? IP { P

(A) Packet Filtering Firewall
(B) Stateful Firewall
(C) Proxy Firewall
(D) Application Firewall

[Source: BCI Subject Test-9.pdf | auto-classified: Topic 4 / Gateway / Firewall]

⚠ auto-ingested — answer needs verification

P12. Which firewall acts as an intermediary between the client and server at the application layer? } ~ P •r 2 B P

(A) Packet Filtering Firewall
(B) Stateful Firewall
(C) Proxy Firewall
(D) Circuit-level Gateway

[Source: BCI Subject Test-9.pdf | auto-classified: Topic 4 / Gateway / Firewall]

⚠ auto-ingested — answer needs verification

P13. Which of the following statements are TRUE? k l B - ƒ u? 1. Switch reduces collision domain. 2. Hub creates a separate collision domain for each port. @ƒ P 3.Router separates broadcast domains. ˆ q

(A) 1
(B) 1 3
(C) 2 3
(D) 1, 2 3

[Source: BCI Subject Test-9.pdf | auto-classified: Topic 4 / Hub / Repeater / Modem]

⚠ auto-ingested — answer needs verification

P14. Which device operates at the Physical layer

(A) Switch
(B) Hub
(C) Router
(D) Bridge

[Source: BCI_SCI Subject Test-7.pdf | auto-classified: Topic 4 / Switch / NIC / Bridge]

⚠ auto-ingested — answer needs verification

P15. Transport layer multiplexing allows: E A f g :h R

(A) Multiple hosts to share one IP
(B) Multiple processes to share one network connection
(C) Multiple routers to share routing table
(D) Domain name resolution

[Source: BCI_SCI Subject Test-7.pdf | auto-classified: Topic 4 / Router / Brouter]

⚠ auto-ingested — answer needs verification

P16. A router performs which of the following? 9 7 - A ?** (1) Packet forwarding (2) Logical addressing (3) Collision detection (4) Routing decision

(A) 1,2,4
(B) 2,3,4
(C) 1,3 only
(D) 1,2,3,4

[Source: BCI_SCI Subject Test-7.pdf | auto-classified: Topic 4 / Router / Brouter]

⚠ auto-ingested — answer needs verification

P17. Consider the following: S1: TTL field decreases at router. S2: MAC address changes at each hop. 7: S1: TTL i S2: MAC E

(A) Only S1
(B) Only S2
(C) Both S1 and S2
(D) Neither S1 nor S2

[Source: BCI_SCI Subject Test-7.pdf | auto-classified: Topic 4 / Router / Brouter]

⚠ auto-ingested — answer needs verification

P18. A router drops packets due to congestion. What is this technique called? R O?**

(A) Delay policing
(B) Redundancy
(C) Random early detection
(D) Reverse flooding

[Source: BCI_SCI Subject Test-7.pdf | auto-classified: Topic 4 / Router / Brouter]

⚠ auto-ingested — answer needs verification

P19. What does MAC address represent? MAC E A ?**

(A) Location of device in network
(B) Unique identity of NIC
(C) Internet connectivity
(D) Routing path

[Source: BCI_SCI Subject Test-7.pdf | auto-classified: Topic 4 / Switch / NIC / Bridge]

⚠ auto-ingested — answer needs verification

P20. Which device separates collision domains

(A) HUB
(B) Switch
(C) Repeater
(D) Bridge

[Source: CN Topic Test 6.pdf | auto-classified: Topic 4 / Switch / NIC / Bridge]

⚠ auto-ingested — answer needs verification

P21. What is a subnet mask used for?Subnet

(A) Encrypt data
(B) Identify default gateway
(C) Identify network and host portion
(D) Manage routing table

[Source: CN Topic Test 6.pdf | auto-classified: Topic 4 / Router / Brouter]

⚠ auto-ingested — answer needs verification

P22. Which of the following devices works only on layer 1

(A) Router
(B) Hub
(C) Switch
(D) Bridge

[Source: CN Topic Test 6.pdf | auto-classified: Topic 4 / Switch / NIC / Bridge]

⚠ auto-ingested — answer needs verification

P23. Which device operates at all 7 layers of OSI model? OSI 7

(A) Router
(B) Hub
(C) Gateway
(D) Switch

[Source: CN Topic Test 6.pdf | auto-classified: Topic 4 / Hub / Repeater / Modem]

⚠ auto-ingested — answer needs verification

P24. BGP is mainly used for which type of routing? BGP

(A) Intra-AS routing
(B) Inter-AS routing
(C) Static routing
(D) Host routing

[Source: CN Topic Test 6.pdf | auto-classified: Topic 4 / Router / Brouter]

⚠ auto-ingested — answer needs verification

P25. What kind of NAT maps one private IP to one public IP? - NAT IP IP

(A) Dynamic NAT
(B) Static NAT
(C) PAT
(D) Port Forwarding

[Source: CN Topic Test 6.pdf | auto-classified: Topic 4 / Gateway / Firewall]

⚠ auto-ingested — answer needs verification

P26. What is the function of NAT in computer networks? NAT

(A) Port forwarding
(B) IP address translation
(C) Bandwidth control
(D) DNS resolution

[Source: CN Topic Test 5.pdf | auto-classified: Topic 4 / Gateway / Firewall]

⚠ auto-ingested — answer needs verification

P27. MPLS operates between which two layers?MPLS

(A) Layer 1 and 2
(B) Layer 2 and 3
(C) Layer 3 and 4
(D) Layer 4 and 7

[Source: CN Topic Test 5.pdf | auto-classified: Topic 4 / Switch / NIC / Bridge]

⚠ auto-ingested — answer needs verification

P28. In specific, if the systems use separate protocols, which one of the following devices is used to link two systems

(A) Repeater
(B) Gateway
(C) Bridge
(D) Hub

[Source: CN Topic Test-01.pdf | auto-classified: Topic 4 / Hub / Repeater / Modem]

⚠ auto-ingested — answer needs verification

P29. Which of the following devices is not a networking device

(A) Hub
(B) Switch
(C) Bridge
(D) None of the these

[Source: CN Topic Test-01.pdf | auto-classified: Topic 4 / Switch / NIC / Bridge]

⚠ auto-ingested — answer needs verification

P30. Which of the following layer protocols are responsible for user and the application programme support such as passwords, resource sharing, file transfer and network management

(A) Layer 2 protocols
(B) Layer 5 protocols
(C) Layer 7 protocols
(D) Layer 10 protocols

[Source: CN Topic Test-02.pdf | auto-classified: Topic 4 / Switch / NIC / Bridge]

⚠ auto-ingested — answer needs verification

P31. Which device connects different networks and routes data between them

(A) Hub
(B) Switch
(C) Router
(D) Bridge

[Source: CN Topic Test-03.pdf | auto-classified: Topic 4 / Switch / NIC / Bridge]

⚠ auto-ingested — answer needs verification

P32. A network ID cannot be assigned to:

(A) A router
(B) A switch
(C) A host device
(D) The entire network

[Source: CN Topic Test-03.pdf | auto-classified: Topic 4 / Switch / NIC / Bridge]

⚠ auto-ingested — answer needs verification

P33. What is the primary function of the MAC sublayer? MAC

(A) Routing
(B) Error checking
(C) Addressing
(D) Channel access

[Source: CN Topic Test-04.pdf | auto-classified: Topic 4 / Router / Brouter]

⚠ auto-ingested — answer needs verification

P34. Which of the following hardware forwards data packets between computer networks?

(A) Switch
(B) Router
(C) Cable
(D) Nibble

[Source: KVS Librarian 2023 Official Paper (via Testbook) | auto-classified: Topic 4 / Switch / NIC / Bridge]

⚠ Official key. Router = internetwork forwarding.

P35. Which device forwards packets between networks by processing the routing information included in the packet?

(A) Firewall
(B) All of these
(C) Router
(D) Bridge

[Source: InstrumentationTools 150 CN MCQs | auto-classified: Topic 4 / Router / Brouter]

⚠ Site key.

P36. Which of the following is NOT a networking device? **(A)** Gateway **(B)** Linux **(C)** Router **(D)** Firewall [Source: InstrumentationTools 150 CN MCQs | auto-classified: Topic 4 / Gateway / Firewall]	⚠ Site key.
P37. Which device operates at Layer 3 of the OSI model? **(A)** Hub **(B)** Switch **(C)** Router **(D)** Repeater [Source: SSC CGL Tier-2 repeated Qs (via Oliveboard) | auto-classified: Topic 4 / Hub / Repeater / Modem]	⚠ Site key.
P38. What is the purpose of ARP (Address Resolution Protocol)? **(A)** To map IP addresses to MAC addresses **(B)** To discover the default gateway **(C)** To resolve domain names to IP addresses **(D)** To establish a secure connection [Source: 30 important CN MCQs (Medium blog bank) | auto-classified: Topic 4 / Gateway / Firewall]	⚠ Site key.
P39. What is the role of a firewall in computer networks? **(A)** Managing IP addresses **(B)** Filtering and controlling network traffic **(C)** Resolving domain names to IP addresses **(D)** Assigning MAC addresses dynamically [Source: 30 important CN MCQs (Medium blog bank) | auto-classified: Topic 4 / Switch / NIC / Bridge]	⚠ Site key.
P40. What is the purpose of a proxy server in networking? **(A)** Managing IP addresses **(B)** Filtering and forwarding network traffic **(C)** Assigning MAC addresses dynamically **(D)** Encrypting network communication [Source: 30 important CN MCQs (Medium blog bank) | auto-classified: Topic 4 / Switch / NIC / Bridge]	⚠ Site key.
P41. Which device operates only in the physical layer? **(A)** Two-layer switch **(B)** Router **(C)** Repeater **(D)** Bridge [Source: Kerala PSC repeated PYQs (via pscarivukal) | auto-classified: Topic 4 / Switch / NIC / Bridge]	⚠ PYQ key (Head of Section Computer Engineering, 2019).

P42. Among these network devices — i. Repeater ii. Hub iii. Switch iv. Router — which are considered intelligent devices? **(A)** i and iv only **(B)** i and ii only **(C)** iii and iv only **(D)** ii and iii only [Source: Kerala PSC repeated PYQs (via pscarivukal) | auto-classified: Topic 4 / Hub / Repeater / Modem]	⚠ PYQ key (University Assistant Main, 2023).
P43. In this network, which of the forwarding tables below ensures that both : - `A` and `B` can exchange packets in both directions - the path from `A` to `B` is the reverse of the path from `B` to `A`. **(A)** New forwarding table for `R3`: **(B)** New forwarding table for `R3`: **(C)** New forwarding table for `R3`: **(D)** New forwarding table for `R3`: [Source: cnp3 MCQ (Bonaventure CN textbook), ftableAdd | auto-classified: Topic 4 / Switch / NIC / Bridge]	⚠ auto-ingested — answer needs verification
P44. 4Htlo0r In a DataEntry Office,thecomputer centers are on Ist and 4th floor.Both the centres are connected three LAN.However the signals take time to travel.Which amongst the 34AN followinghardware canimprove the data transmission? **(A)** Modem **(B)** Repeater **(C)** 1s+ flor **(D)** Router [Source: Random Test- Rajasthan PYQ.pdf (OCR), p2 | auto-classified: Topic 4 / Hub / Repeater / Modem]	⚠ OCR-parse staging: verify options/answer at curation (see src page).
P45. is a network security system designed to protect a trustedprivatenetwork from an untrusted outside network to which it is connected. Bridge Cookies Firewall [Source: Random Test- Rajasthan PYQ.pdf (OCR), p6 | auto-classified: Topic 4 / Switch / NIC / Bridge]	⚠ OCR-parse staging: verify options/answer at curation (see src page). OPTIONS INCOMPLETE in OCR — re-check source page.
P46. 93.Network devices used atphysical layer. Gateway,Bridge Router,Repeaier Hub,Switch Repeater,Hub [Source: Random Test- Rajasthan PYQ.pdf (OCR), p12 | auto-classified: Topic 4 / Hub / Repeater / Modem]	⚠ OCR-parse staging: verify options/answer at curation (see src page). OPTIONS INCOMPLETE in OCR — re-check source page.
P47. which of the following 9. are network connecting devices 1. repeater 2. Bridge 3. hub4. bluetooth **(A)** 1,2 and 3 **(B)** 1,2 and 4 (o) 2,3 and 4 (**(C)** 1,3 and 4 [Source: Unit Test -2 Solutions.pdf (OCR), p11 | auto-classified: Topic 4 / Hub / Repeater / Modem]	⚠ OCR-parse staging: verify options/answer at curation (see src page).

P48. 7. In early ARPANET, which routing method was used: (a/Distance vector routing - Locl Qalabax

(A) Link state routing →→Cleba Qalahav

(B) Least cost routing

(C) Flooding

[Source: Unit Test-4 Solution.pdf (OCR), p8 | auto-classified: Topic 4 / Router / Brouter]

⚠ OCR-parse staging: verify options/answer at curation (see src page).

P49. 14. Consider the following statements about the functionality of an IP based router. QX A router does modify during packets forwarding. JI. It is necessary for a router to i implement routing protocol. WI. A router should 1 reassemble IP f fragments if the MTU of the outgoing link is larger than the size of the incoming packet. Which of the above statements is/ are TRUE?

(A) I and 1I only

(B) I only

(C) 1I and 11I only 11 only

[Source: Unit Test-4 Solution.pdf (OCR), p15 | auto-classified: Topic 4 / Router / Brouter]

⚠ OCR-parse staging: verify options/answer at curation (see src page).

P50. 21. is a device that combines transmission from various I/O devices into one line

(A) Modifier Multiplexer

(B) Concentrator (d) Gateway

[Source: Unit Test-4 Solution.pdf (OCR), p22 | auto-classified: Topic 4 / Gateway / Firewall]

⚠ OCR-parse staging: verify options/answer at curation (see src page).

P51. 23. File server is a which is used to store files received from various client machines

(A) Specialized computer system

(B) Modem

(C) Router

(D) Transponder

[Source: Unit Test-4 Solution.pdf (OCR), p24 | auto-classified: Topic 4 / Hub / Repeater / Modem]

⚠ OCR-parse staging: verify options/answer at curation (see src page).

P52. 26. Dynamic routing protocol enable routers to:

(A) Dynamically discover and maintain router

(B) Distribute routing updates to other routers

(C) Reach agreement with other routers about the network topology

(D) All of the above

[Source: Unit Test-4 Solution.pdf (OCR), p27 | auto-classified: Topic 4 / Router / Brouter]

⚠ OCR-parse staging: verify options/answer at curation (see src page).

P53. Devices on one network can communicate with devices on another network via:

(A) File server

(B) Utility server

(C) Printer server

(D) Gateway

[Source: SSC CGL Internet & Networking (via EduRev bank) | auto-classified: Topic 4 / Types of Networks]

⚠ Site key.